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System boundary

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System: Inspection Report Routing System

Inside the boundary

The Inspection Report Routing System

Outside the boundary

Flight control, physical inspections, and repair work

Boundary flows

One row per information flow. Repeat an entity if it has more than one flow.

External entity	Direction (into / out of system)	Data flow name	Source
Field Team	In	Inspection Report	W06-S01
Field Team	Out	Validation Note	W06-S01
Electrical	Out	Work Item	W06-S01
Drainage	Out	Work Item	W06-S01
Structures	Out	Work Item	W06-S01

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Information movement

Processes: 1 Receive Report; 2 Check Required Fields; 3 Resolve Destination; 4 Send Work Item. Stores: D1 Report Register; D2 Routing Table. Use the flow rows to connect these and the external teams from page 1. Eleven flow rows are provided.

From (entity / process / store)	To (entity / process / store)	Data flow name	Source
Field Team (Entity)	Receive Report (Process)	Inspection Report	W06-S01
Receive Report (Process)	D1 Report Register (Store)	Submitted Report	W06-S01
D1 Report Register (Store)	Check Required Fields (Process)	Received Report	W06-S01
Check Required Fields (Process)	Field Team (Entity)	Validation Note	W06-S01
Check Required Fields (Process)	Resolve Destination (Process)	Accepted Report	W06-S01
D2 Routing Table (Store)	Resolve Destination (Process)	Routing Entries	W06-S01
Resolve Destination (Process)	Send Work Item (Process)	Resolved Destination	W06-S01
Send Work Item (Process)	Electrical (Entity)	Work Item	W06-S01
Send Work Item (Process)	Drainage (Entity)	Work Item	W06-S01
Send Work Item (Process)	Structures (Entity)	Work Item	W06-S01
Send Work Item (Process)	D1 Report Register (Store)	Delivery Record	W06-S01

Balance check:

The Field Team and individual departments to don't directly interact with the stores, thus meaning that they remain outside of the system, and can only interact through the processes. This means that the Information movement is compatible with the Boundary flows because the data flowing between them is the same in both models, and the external entities don't interact with the stores.

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Data definitions and finding

Field	Meaning	Example	Allowed value / format	Source
report_id	A uniquely numbered report	CJ-609	CJ- followed by three digits.	W06-S02
asset_id	Reference for the inspected asset	SL-293	Two-letter prefix, then three digits	W06-S02
asset_type	The name of the asset	DRAIN	STREETLIGHT, DRAIN, or BRIDGE	W06-S02
finding_code	The problem	BLOCKED	LAMP_OUT, BLOCKED, SURFACE_DAMAGE	W06-S02
priority	The urgency of the problem	URGENT	ROUTINE or URGENT	W06-S02
captured_at	Date and time problem was recognized	2026-10-31T11:50:00Z	UTC capture time; YYYY-MM-DDTHH:MM:SSZ	W06-S02
photo_ref	Image or visual information	IMG-609	IMG- foll0wed by three digits	W06-S02
survey_area	Area where the survey took place	NORTH	NORTH, CENTRAL or SOUTH	W06-S02

Supported finding

CJ-603 recounts that surface damage has occurred on a bridge in the Central survey area; the request is marked as BRIDGE. According to W06-S03, Structures should be taking lead on any asset types with BRIDGEs, yet W06-S04 shows that Electrical has instead been the receiving department.

Unanswered question:

Why are BRIDGE asset_type fields being defaulted to the Electrical department?